

## PITLYAR, Russian Federation

WMO: 234310

Lat: 65.8414N

Lon: 65.9125E

Elev: 54

StdP: 14.67

Time Zone: 5.00 (E05)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -45.0      | -40.8 | -49.3                         | 0.3 | -44.1 | -44.9 | 0.4 | -40.2 | 26.0                  | 9.5  | 22.3 | 16.3 | 3.2                   | 90   | 0.768 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.0                   | 81.2            | 65.9 | 76.3 | 64.0 | 72.2 | 61.7 | 67.7                | 77.2 | 65.4 | 74.5 | 63.1 | 70.9 | 7.0                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.4                            | 90.6 | 73.0 | 61.8 | 82.6 | 69.7 | 59.5 | 75.9 | 67.4 | 32.1          | 77.3 | 30.4 | 74.5 | 28.6 | 71.0 | 75.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 23.3          | 20.4 | 17.2 | DB                         | -47.4 | 83.8               | 3.8 | 5.3                                                | -50.2 | 87.6       | -52.4 | 90.8       | -54.6 | 93.7       | -57.3 | 97.6 | (4) |
| (5)               |      |      | WB                         | -47.0 | 69.7               | 3.2 | 2.9                                                | -49.3 | 71.8       | -51.1 | 73.5       | -52.9 | 75.2       | -55.2 | 77.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 24.7   | -10.7 | -5.6  | 11.9  | 20.5  | 34.8  | 53.5  | 62.0  | 53.8  | 43.3  | 27.5  | 5.8   | -2.5  |  |  |
|                                                                        | DBStd       | 27.04  | 16.57 | 16.89 | 14.33 | 11.52 | 9.22  | 9.41  | 7.98  | 6.26  | 6.70  | 9.16  | 14.51 | 17.64 |  |  |
|                                                                        | HDD50       | 9960   | 1881  | 1556  | 1180  | 884   | 484   | 71    | 5     | 31    | 218   | 697   | 1327  | 1626  |  |  |
|                                                                        | HDD65       | 14787  | 2346  | 1976  | 1645  | 1334  | 937   | 360   | 158   | 349   | 652   | 1162  | 1777  | 2091  |  |  |
|                                                                        | CDD50       | 729    | 0     | 0     | 0     | 0     | 12    | 175   | 376   | 150   | 16    | 0     | 0     | 0     |  |  |
|                                                                        | CDD65       | 81     | 0     | 0     | 0     | 0     | 0     | 14    | 64    | 3     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 645    | 0     | 0     | 0     | 0     | 0     | 91    | 542   | 12    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 153    | 0     | 0     | 0     | 0     | 0     | 10    | 143   | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 7.4    | 5.9   | 5.9   | 7.9   | 8.0   | 8.6   | 8.8   | 7.4   | 7.6   | 7.8   | 8.0   | 6.3   | 6.0   |  |  |
|                                                                        | PrecAvg     | 34.4   | 2.2   | 1.9   | 2.7   | 2.8   | 2.8   | 3.1   | 3.1   | 3.4   | 2.8   | 4.1   | 3.0   | 2.5   |  |  |
|                                                                        | PrecMax     | 40.8   | 3.5   | 3.1   | 5.8   | 5.7   | 4.5   | 4.6   | 6.3   | 6.7   | 5.7   | 5.9   | 5.1   | 3.8   |  |  |
| (15) Precipitation                                                     | PrecMin     | 29.8   | 1.5   | 0.7   | 0.8   | 1.1   | 1.2   | 1.3   | 0.6   | 1.5   | 0.9   | 1.5   | 1.9   | 1.5   |  |  |
|                                                                        | PrecStd     | 2.4    | 0.5   | 0.6   | 0.8   | 0.9   | 0.9   | 0.9   | 1.3   | 1.2   | 1.0   | 0.9   | 0.7   | 0.6   |  |  |
|                                                                        | DB          | 28.9   | 30.2  | 39.4  | 45.1  | 67.4  | 81.4  | 88.0  | 75.4  | 65.6  | 49.5  | 35.3  | 31.5  |       |  |  |
|                                                                        | MCWB        | 27.7   | 28.7  | 33.9  | 38.4  | 56.3  | 64.9  | 68.1  | 64.1  | 57.1  | 45.5  | 33.8  | 29.9  |       |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2%          | DB     | 23.3  | 25.6  | 35.8  | 42.2  | 60.9  | 76.3  | 84.2  | 70.9  | 59.7  | 45.0  | 31.1  | 27.5  |  |  |
|                                                                        |             | MCWB   | 22.2  | 24.5  | 31.9  | 36.7  | 50.8  | 61.5  | 67.3  | 60.9  | 53.3  | 42.4  | 29.7  | 26.3  |  |  |
|                                                                        | 5%          | DB     | 18.1  | 21.9  | 33.4  | 39.2  | 54.0  | 72.5  | 80.4  | 68.0  | 56.0  | 41.4  | 28.5  | 24.4  |  |  |
|                                                                        |             | MCWB   | 17.2  | 20.9  | 30.5  | 35.3  | 46.6  | 60.0  | 66.2  | 59.3  | 51.5  | 38.9  | 27.5  | 23.6  |  |  |
|                                                                        | 10%         | DB     | 13.2  | 17.5  | 31.2  | 36.6  | 47.8  | 68.2  | 76.1  | 65.1  | 52.8  | 39.0  | 24.7  | 21.1  |  |  |
|                                                                        |             | MCWB   | 12.3  | 16.5  | 28.9  | 33.3  | 42.0  | 58.3  | 64.8  | 58.0  | 49.4  | 37.4  | 23.7  | 20.3  |  |  |
|                                                                        | 0.4%        | WB     | 27.8  | 29.0  | 35.1  | 40.3  | 56.5  | 67.3  | 72.3  | 66.1  | 58.1  | 46.7  | 33.9  | 30.5  |  |  |
|                                                                        |             | MCDB   | 28.7  | 30.2  | 38.7  | 42.6  | 65.4  | 76.8  | 82.5  | 73.6  | 64.9  | 47.7  | 35.5  | 31.3  |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 22.1  | 24.5  | 32.6  | 37.3  | 51.5  | 64.3  | 69.4  | 63.1  | 54.5  | 42.2  | 29.8  | 26.5  |  |  |
|                                                                        |             | MCDB   | 23.1  | 25.3  | 35.1  | 41.4  | 60.4  | 72.5  | 78.0  | 68.2  | 58.3  | 44.9  | 30.7  | 27.5  |  |  |
|                                                                        | 5%          | WB     | 17.1  | 21.1  | 30.7  | 35.2  | 47.1  | 61.7  | 67.4  | 61.0  | 52.2  | 39.4  | 27.5  | 23.7  |  |  |
|                                                                        |             | MCDB   | 17.9  | 22.0  | 33.1  | 38.8  | 54.2  | 70.3  | 76.9  | 66.0  | 55.1  | 41.1  | 28.5  | 24.5  |  |  |
|                                                                        | 10%         | WB     | 12.4  | 16.6  | 28.9  | 33.4  | 42.2  | 59.0  | 65.4  | 58.9  | 49.7  | 37.2  | 23.8  | 20.4  |  |  |
|                                                                        |             | MCDB   | 13.4  | 17.6  | 31.4  | 36.4  | 47.0  | 67.8  | 74.9  | 64.2  | 52.4  | 38.6  | 24.7  | 21.1  |  |  |
|                                                                        | MDBR        | 14.3   | 14.5  | 16.5  | 16.8  | 13.0  | 14.7  | 16.0  | 13.7  | 10.4  | 8.5   | 13.2  | 13.6  |       |  |  |
|                                                                        | 5% DB       | MCDBR  | 17.5  | 13.5  | 12.3  | 14.8  | 20.7  | 19.6  | 22.1  | 18.8  | 15.5  | 10.3  | 13.5  | 11.6  |  |  |
| (36) Mean Daily Temperature Range                                      |             | MCWBR  | 17.1  | 12.8  | 9.8   | 10.8  | 12.5  | 10.1  | 9.1   | 11.1  | 10.1  | 7.7   | 12.5  | 11.4  |  |  |
|                                                                        | 5% WB       | MCDBR  | 17.2  | 13.6  | 12.4  | 14.1  | 19.8  | 17.9  | 19.5  | 15.6  | 13.7  | 8.9   | 13.1  | 11.8  |  |  |
|                                                                        |             | MCWBR  | 16.9  | 12.9  | 10.3  | 10.5  | 12.6  | 10.5  | 9.4   | 10.8  | 9.6   | 7.3   | 12.4  | 11.5  |  |  |
|                                                                        | taub        | 0.207  | 0.264 | 0.312 | 0.336 | 0.353 | 0.369 | 0.408 | 0.376 | 0.325 | 0.285 | 0.204 | 0.214 |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taud        | 1.886  | 2.043 | 2.047 | 2.099 | 2.276 | 2.431 | 2.310 | 2.373 | 2.482 | 2.255 | 1.925 | 1.898 |       |  |  |
|                                                                        | Ebn at Noon | 71     | 185   | 231   | 259   | 268   | 267   | 250   | 247   | 238   | 180   | 69    | 8     |       |  |  |
|                                                                        | Edh at Noon | 6      | 19    | 31    | 38    | 35    | 31    | 34    | 29    | 21    | 15    | 6     | 2     |       |  |  |
|                                                                        | RadAvg      | 29     | 218   | 711   | 1343  | 1701  | 1643  | 1654  | 1073  | 585   | 216   | 52    | 6     |       |  |  |
| (44) All-Sky Solar Radiation                                           | RadStd      | 2      | 22    | 47    | 83    | 114   | 128   | 174   | 90    | 58    | 29    | 6     | 0     |       |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page